

A Browser-Native Pipeline for Faithful Paper Translation

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Abstract: Academic paper translation must preserve technical meaning, immutable visual assets, and the source layout while allowing natural pagination in the target language.

Keywords: document translation, layout inheritance, semantic alignment

1 Introduction

Modern systems papers combine prose with figures, tables, formulas, citations, and code. A faithful workflow translates the prose but leaves every immutable technical object unchanged [1].

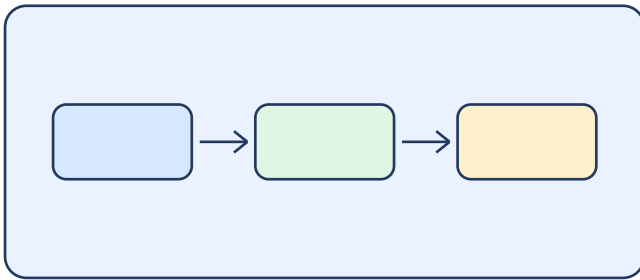


Figure 1: Browser translation pipeline

The browser stores recoverable project state locally. Translation requests contain only the selected text blocks and never include the user's secret in exported project packages.

2 Layout Inheritance

Single-column regions remain single-column, and double-column regions remain double-column. Mixed papers are processed region by region, while the Chinese page count may extend naturally.

3 Semantic Alignment

English and Chinese do not share identical grammar. The reader therefore uses stable semantic groups instead of pretending that every sentence has a strict one-to-one counterpart.

$$latency = parse + translate + compile \quad (1)$$

Equation (1) is an immutable object. Its variables, symbols, subscripts, and number must be preserved exactly, even when adjacent explanatory text is translated.

4 Completion Criteria

A run is complete only after translation validation, deterministic composition, PDF compilation, alignment geometry resolution, and final quality checks all succeed.

References

[1] A. Example and B. Example, Reliable Browser Document Processing, 2026.